

Mark Park

Lead Software Engineer

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SUMMARY

Lead Software Engineer with 4+ years building and running regulated-finance systems at Franklin Templeton, full stack across Java/Spring microservices and Angular UIs (Azure, Drools, Python tooling) — led a 5-engineer rebuild of the UHNW client-onboarding platform from POC to production, owned live-market transaction processing (\$106B+ lifetime), and established a TDD-first agentic development practice on GitHub Copilot CLI adopted across three product teams.

TECHNICAL SKILLS

Languages: Java 8/17/21 (led multi-version migrations), Python (internal tooling and automation), SQL, TypeScript, JavaScript (ES6), C++

Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Cloud (Eureka, Feign Client, API Gateway), Spring Batch, Spring WebFlux, Hibernate/JPA, Lombok, REST APIs, gRPC, OpenAPI/Swagger, Microservices, Domain-Driven Design, Drools (BRMS), Camunda, SAGA Choreography, Event-Driven Architecture

Front-End & Full Stack: Angular (production UI development alongside Spring services), React, TypeScript

Azure: Service Bus, Event Grid, Cosmos DB, Functions, Key Vault, Blob Storage, API Management, Entra ID (Azure AD), Application Insights, Log Analytics (KQL), File Directory System

AWS & Infrastructure: AWS (EC2, S3, ECS, Elastic Beanstalk), Docker, RabbitMQ, Redis

DevOps & Observability: CI/CD, GitHub Actions, Azure Pipelines, Jenkins, SonarQube, Datadog, Maven, Gradle, Git/GitHub

Security & Compliance: OAuth2/OIDC, Azure Entra ID, OFAC screening and KYC workflows, government audit trails

Databases: MSSQL (stored procedures), Azure Cosmos DB, Snowflake, MySQL, MongoDB

AI-Assisted Development — Tools: GitHub Copilot (IntelliJ, premium tier / multi-model, Copilot CLI), GitHub MCP (Model Context Protocol), Claude Code (self-taught on personal time; parallel git-worktree workflows)

AI-Assisted Development — Practice & Leadership: Test-Driven Development with AI agents, agentic code review, agent-authored pull requests, prompt and loop engineering, authored reusable copilot-instructions.md standards and custom CLI commands, led AI adoption across three product teams

Concurrency: Multithreading, custom thread pool executors, optimistic concurrency control, retryable services, deadlock resolution

Testing & Process: Unit and integration testing (JUnit, Mockito, JaCoCo), Postman, code review, technical documentation, root-cause analysis, full SDLC, Jira (incl. Copilot CLI Jira integration), Agile/Scrum, Kanban

Working Knowledge: Kafka, Kubernetes, PostgreSQL, Kotlin, Go; familiar with GraphQL

PROFESSIONAL EXPERIENCE

Brillio — Client: Franklin Templeton (Fiduciary Trust) | Remote (Eastern hours) — New York, NY

Contract Software Engineer (May 2022 – Jun 2025) → Software Development Lead, full-time Brillio employee (Jun 2025 – Present)

Embedded 4+ years on Franklin Templeton engineering teams; performance earned a two-year contract extension, conversion to full-time Brillio employee, and promotion to Software Development Lead. Recognized by directors, managers, architects, and lead engineers from other teams for the successful releases of CashApp, SWP Forms, and Account Opening Platform 2.0. Based in California, worked full Eastern-time hours (6 AM PT start) throughout; open to NY hybrid or CA office.

Lead Software Engineer — Account Opening Platform 2.0 (UHNW Client Onboarding) | Aug 2024 – Present

- Promoted to lead the backend engineering effort on the recommendation of senior architects and management; led a team of up to 5 engineers rebuilding a heavily criticized legacy client-facing platform from POC to MVP to production, serving 500–1,000 daily active users.
- Platform's production release was recognized by leadership as a successful turnaround of a system users and stakeholders had previously criticized, validating both the rebuild's architecture and the team's execution.
- Architected the platform's two core microservices along clear domain boundaries: a Drools-based rules engine governing all UI field behavior, and a secure persistence service acting as the sole authorized gateway to internal Franklin Templeton systems.
- Established a TDD-first agentic delivery workflow on GitHub Copilot CLI: test suites are written from business requirements first, AI agents implement against them, and every change is gated by agent-driven code review and agent-authored pull requests via GitHub MCP — under a standing rule that existing functional tests are never broken.

- Authored the reusable AI standards adopted across teams: a shared copilot-instructions.md (architecture, coding standards, code-comment conventions, and structured commit messages keyed to Jira ID, causation, and rationale for future searchability), custom CLI commands including Jira integration, and a Python tool that aligns database schemas and data across environments, adopted by other projects.
- Grew from backend lead into a full-stack engineer on the platform, delivering Angular UI features alongside the Spring Boot services, and built Python automation projects used as shared internal tooling.
- Led AI adoption across three product teams: ran knowledge-transfer sessions on prompt and loop engineering that onboarded all 5 AOP 2.0 engineers plus the CashApp and SWP Forms leads; drove GitHub Copilot usage at the firm's Jan 2026 rollout — top 5% firm-wide on cost-efficiency, with 30K+ lines of AI-assisted code accepted into production per Franklin Templeton's Copilot metrics.
- Self-taught agentic workflows (parallel git-worktree agents, prompt and loop engineering) on Claude Code on personal time, then translated them onto the firm-approved GitHub Copilot CLI as soon as it was enabled, retaining ownership of architecture and review.
- Led the platform's Java 17→21 migration ahead of the MVP release, completing the upgrade with minimal disruption to the delivery timeline.
- Sole backend engineer on-call during live-market (production) hours — monitored and maintained services in real time while concurrently developing features for upcoming releases.
- Automated over 60% of previously manual onboarding and KYC logic and improved responsiveness by 60–80% over the legacy platform.
- Built a generic recursive Cosmos DB save utility on the Patch API supporting arbitrary object structures at 5–10 ms response times, working around a known SDK upsert limitation.
- Mentored and managed a summer intern new to enterprise Java, guiding them to ship several production APIs for database persistence and workflow logic.

Backend Engineer — SWP Forms (Systematic Withdrawal Plan) | Aug 2024 – Dec 2025 (concurrent with AOP 2.0)

- Hand-picked by the lead architect to build a standalone Drools rules-engine microservice from concept through MVP; cut processing time for 500+ business rules from several seconds to 50–100 ms using fact injection, agenda groups, and salience-based prioritization.
- Delivered the MVP in 3 months; the service was later extended to power a second production application.
- After handoff, served as the solution architect's designated assistant: onboarded and trained the engineer taking over the application and resolved production issues through Dec 2025.

Backend Developer — CashApp (Franklin Templeton internal transaction platform) | May 2022 – Aug 2024

- Mentored by a 20-year architect on a 3-person team; became a primary contributor on core transaction-processing design across the 18-month POC-to-MVP build and subsequent enhancement releases.
- Served as one of two primary production maintainers, personally overseeing live-market-hours processing of time- and data-sensitive transactions collectively worth tens of millions of dollars daily; the pipeline's strict timing windows meant any missed cutoff or incomplete data forced a restart or rolled settlement to the next business day — an outcome never permitted to occur, with coverage coordinated (down to staggered lunches) so one maintainer was always available, often in real-time coordination with teams across the firm.
- Drove major post-MVP enhancements: bulk transaction processing, deprecation of legacy processing in favor of FIS Global Plus posting — a multi-currency portfolio accounting and trust custody platform used to enter and settle trades, cash receipts, and disbursements — and integration of Azure-based check-reading AI.
- Owned the end-to-end financial processing lifecycle for \$10M+ in daily transactions (\$106B+ lifetime): bank confirmations, OFAC compliance scans, and government audit-trail processing.
- Designed an event-driven architecture on Azure Event Grid with message queues powering 20+ heavy scheduled jobs — several on 5-minute cycles, including OFAC scanning and posting transactions to banks.
- Eliminated production deadlock failures — from 2–5 incidents per day to zero — by refactoring the application workflow with an optimistic concurrency control layer, retryable services, and custom thread pool executors.
- Implemented SAGA choreography over RabbitMQ for cross-service transactions, later migrating messaging to Azure Service Bus.
- Led technical design discussions across cross-functional groups of 20–30 with product owners and business analysts.
- Subsequently selected to assist the CashApp team's Java 8→21 upgrade, applying migration experience from Account Opening Platform 2.0 to modernize a long-lived production codebase across multiple major language versions.

EDUCATION

San Diego State University — Bachelor of Science, Computer Science